

磁共振影像研究

Magnetic Resonance Imaging (MRI)

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生物医学工程系

课程内容

- 磁共振成像扫描仪历史、成像原理、构造、特点、安全注意事项
- 临床与科研应用
- 实验室参观和亲身体验磁共振成像

Take Home Message:

- MRI is safe
- MRI is fun



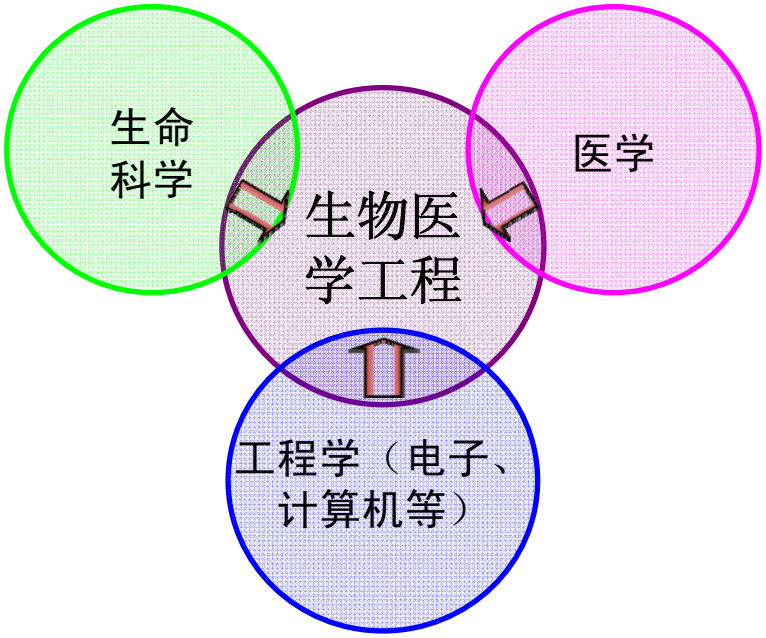
你我关心健康

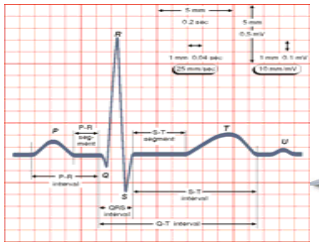


5:30AM, 门诊大厅前

生物医学工程

生物医学工程是工程学、生命科学和医学紧密结合的交叉学科，它将工程学原理用于医学、生物学和健康领域，从分子、细胞、组织、器官、系统、人体等多层次发展这些领域的新概念和新知识，研制用于预防、诊断、治疗疾病及促进健康的创新性医疗设备、生物制剂、生物材料、生物过程、植入设备等。生物医学工程是极具前景的朝阳学科，将带动整个工程学科、医学学科及相关产业的发展。





心电图



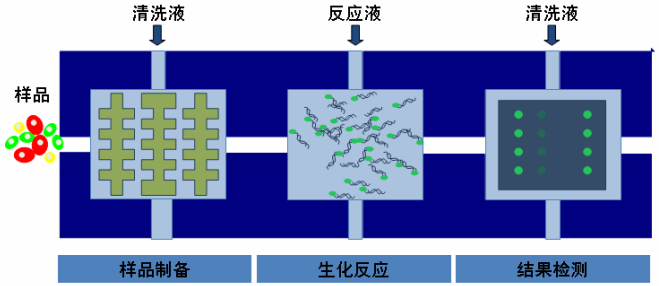
医用监护仪



磁共振成像



多通道脑电检测

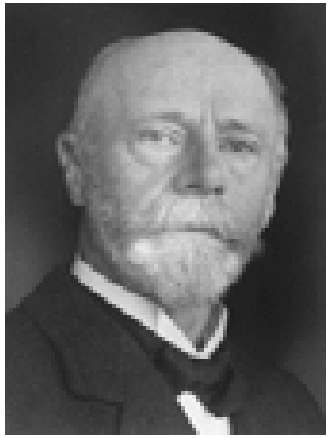


用于医学检验的微流控芯片

与生物医学工程有关的诺贝尔奖



W C Roentgen
1900年物理奖
X光



W Einthoven
1924年生理医学奖
心电图



I Rabi
1944年物理奖
磁共振

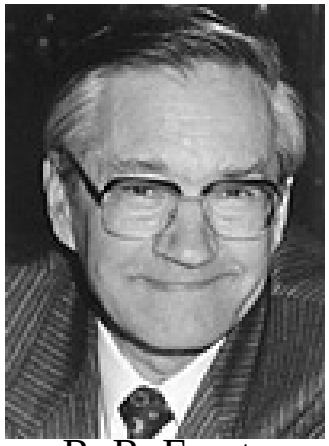


F Bloch & E Purcell
1952年物理奖
磁共振





G N Hounsfield & A M Cormack
1979年生理医学奖
X-CT



R. R. Ernst
1991年化学奖
磁共振





P C Lauterbur & P Mansfield
2003年生理医学奖
磁共振

生物医学影像

- X-ray
- CT
- US
- MRI
- PET
- OCT
- …




Neusoft

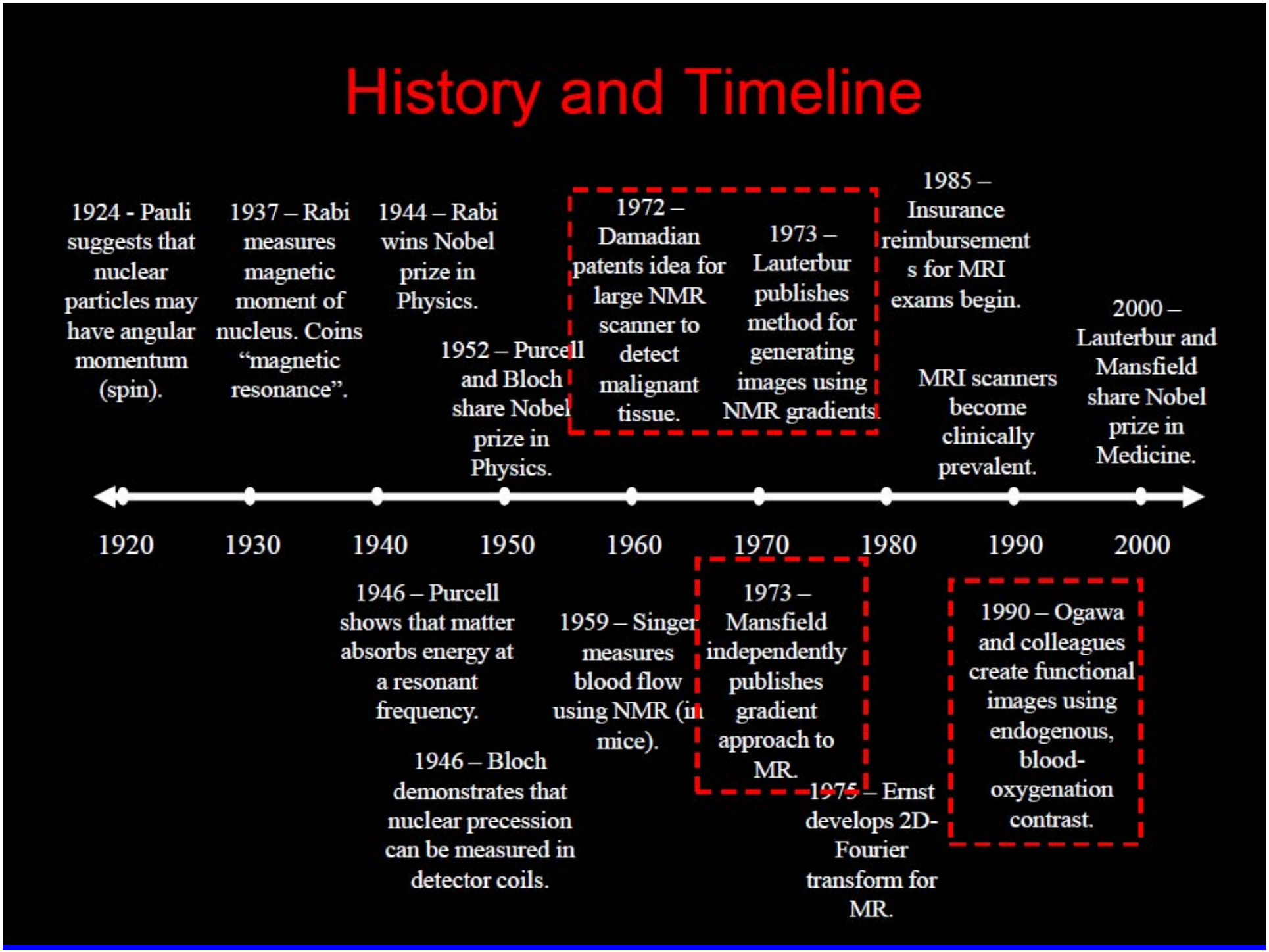

ALLEGRO
ADAC
Laboratories


PHILIPS

MRI

磁共振发展历史

History and Timeline



History of MRI

第一幅 NMR 图像

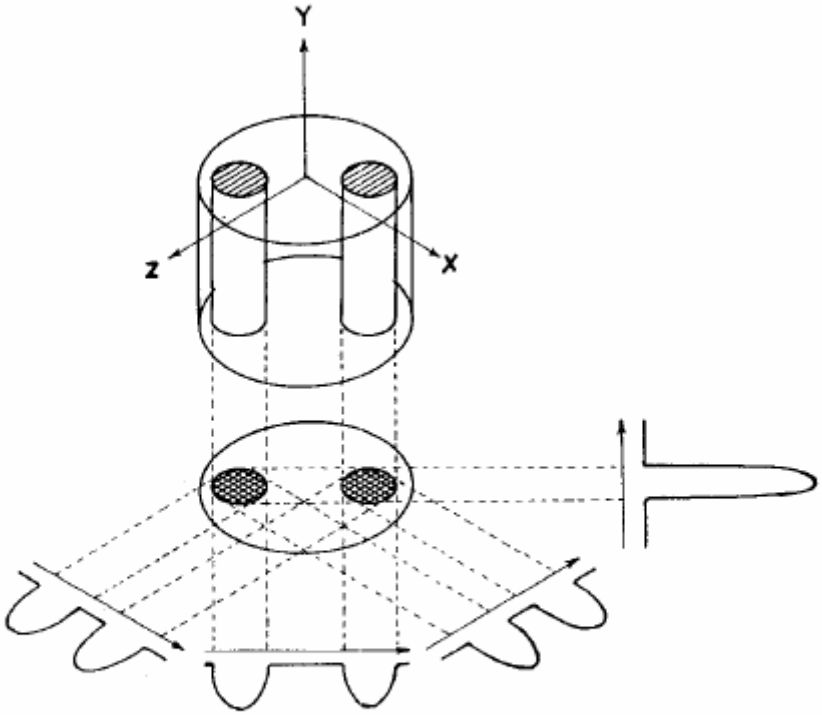


Fig. 1 Relationship between a three-dimensional object, its two-dimensional projection along the Y-axis, and four one-dimensional projections at 45° intervals in the XZ-plane. The arrows indicate the gradient directions.

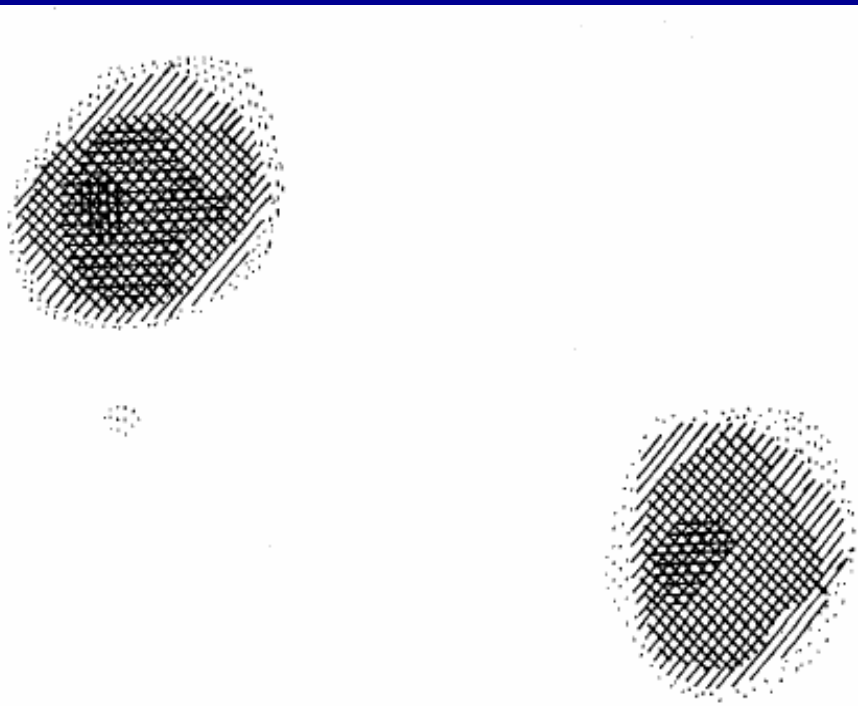
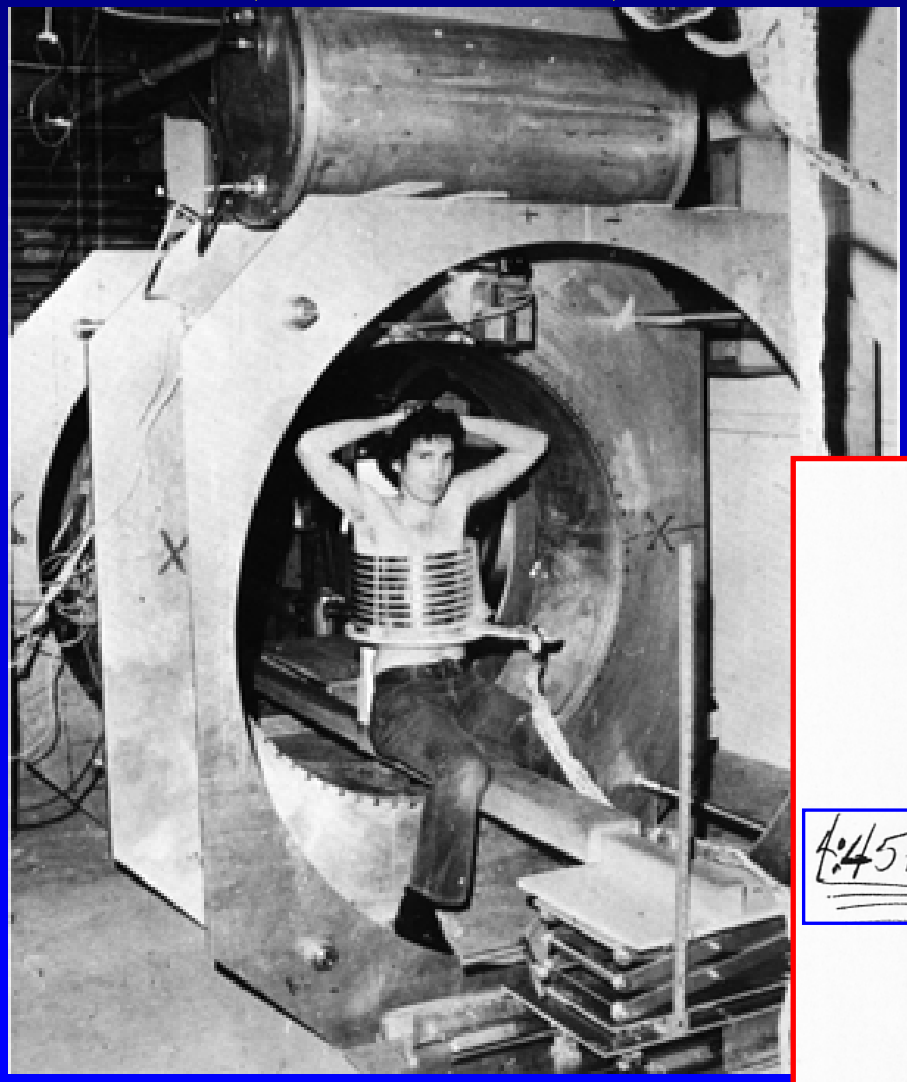


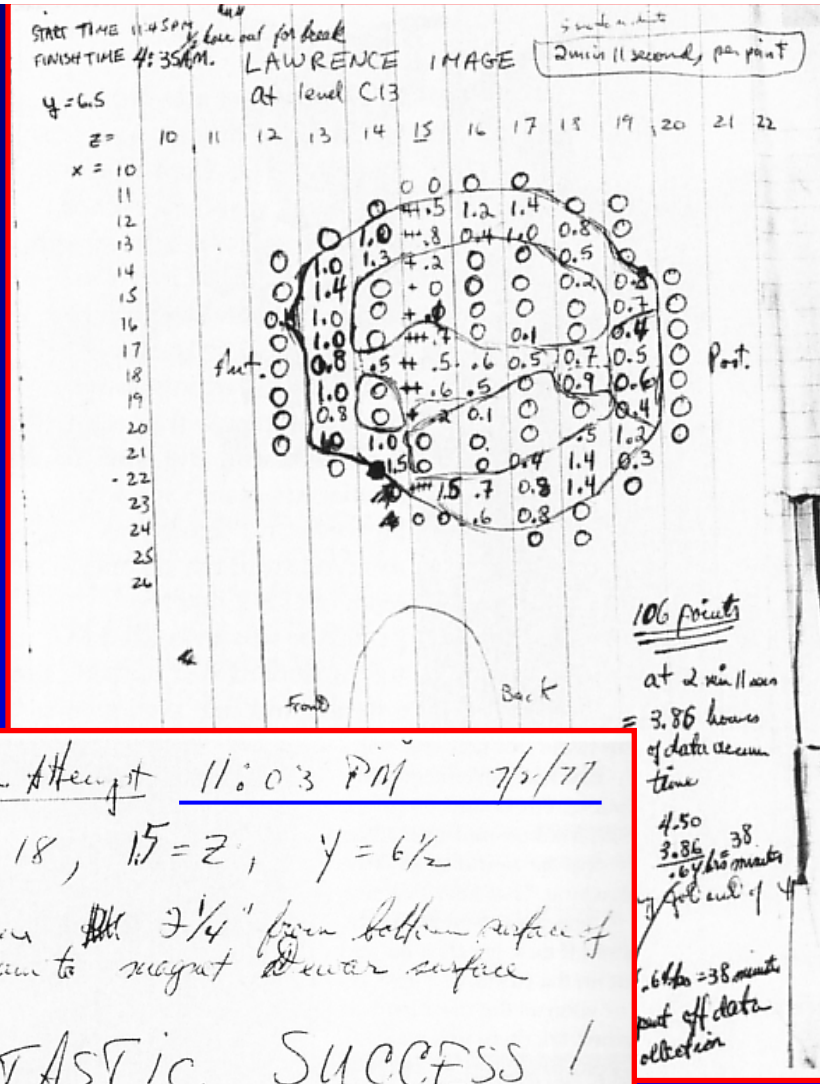
Fig. 2 Proton nuclear magnetic resonance zeugmatogram of the object described in the text, using four relative orientations of object and gradients as diagrammed in Fig. 1.

Lauterbur, P.C. (1973). Image formation by induced local interaction: Examples employing nuclear magnetic resonance. *Nature*, 242, 190-191.

History of MRI

早期MR图像 (Damadian)





Human Attempt 11:03 PM 7/2/77

X = 18, 15 = Z, Y = 6½

Beam 7¼" from bottom surface of beam to magnet center surface

FANTASTIC SUCCESS!

4:45 A.M.

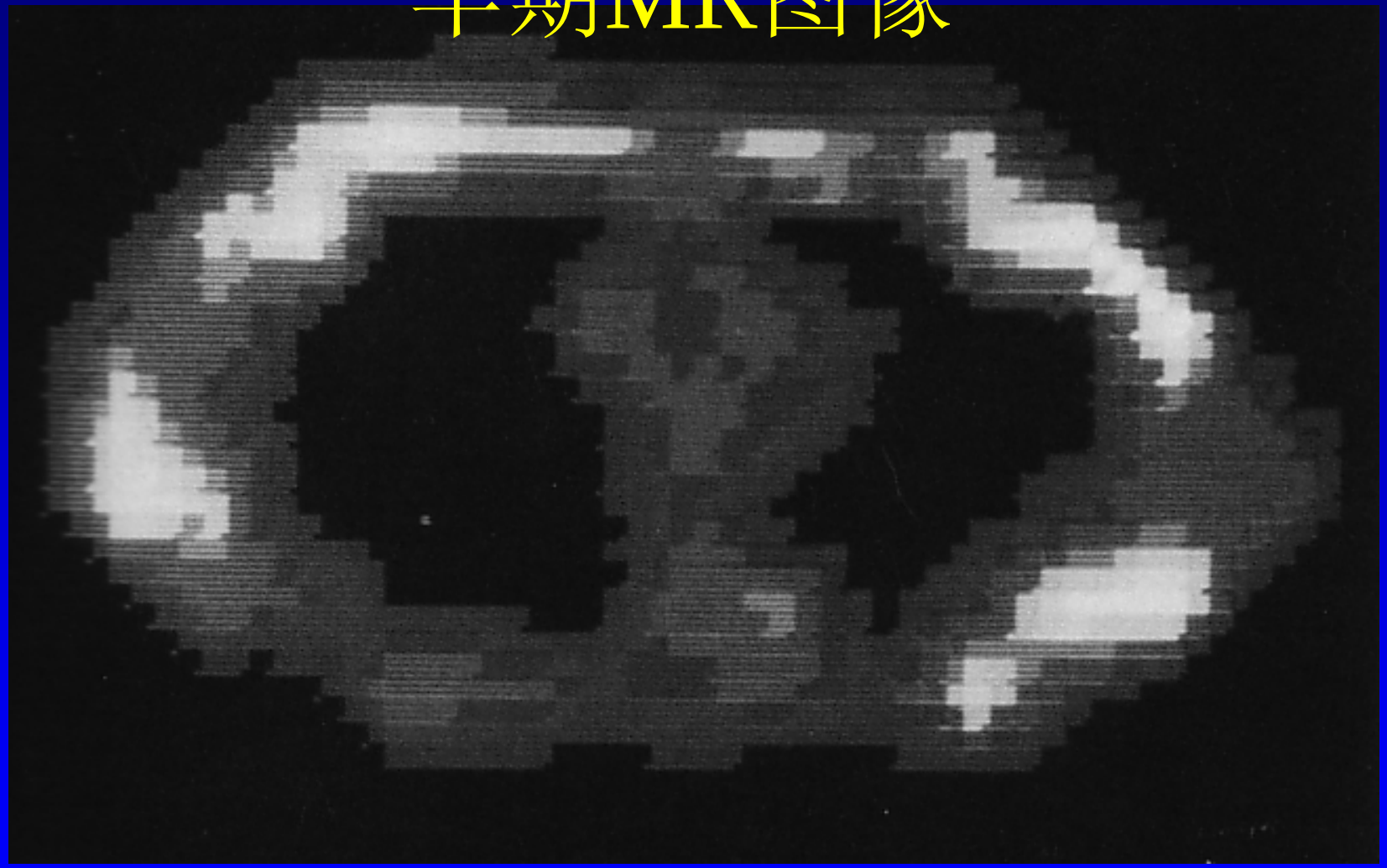
First Human Image

Complete in Amazing Detail

Showing Heart
Lungs
Vertebra
Musculature

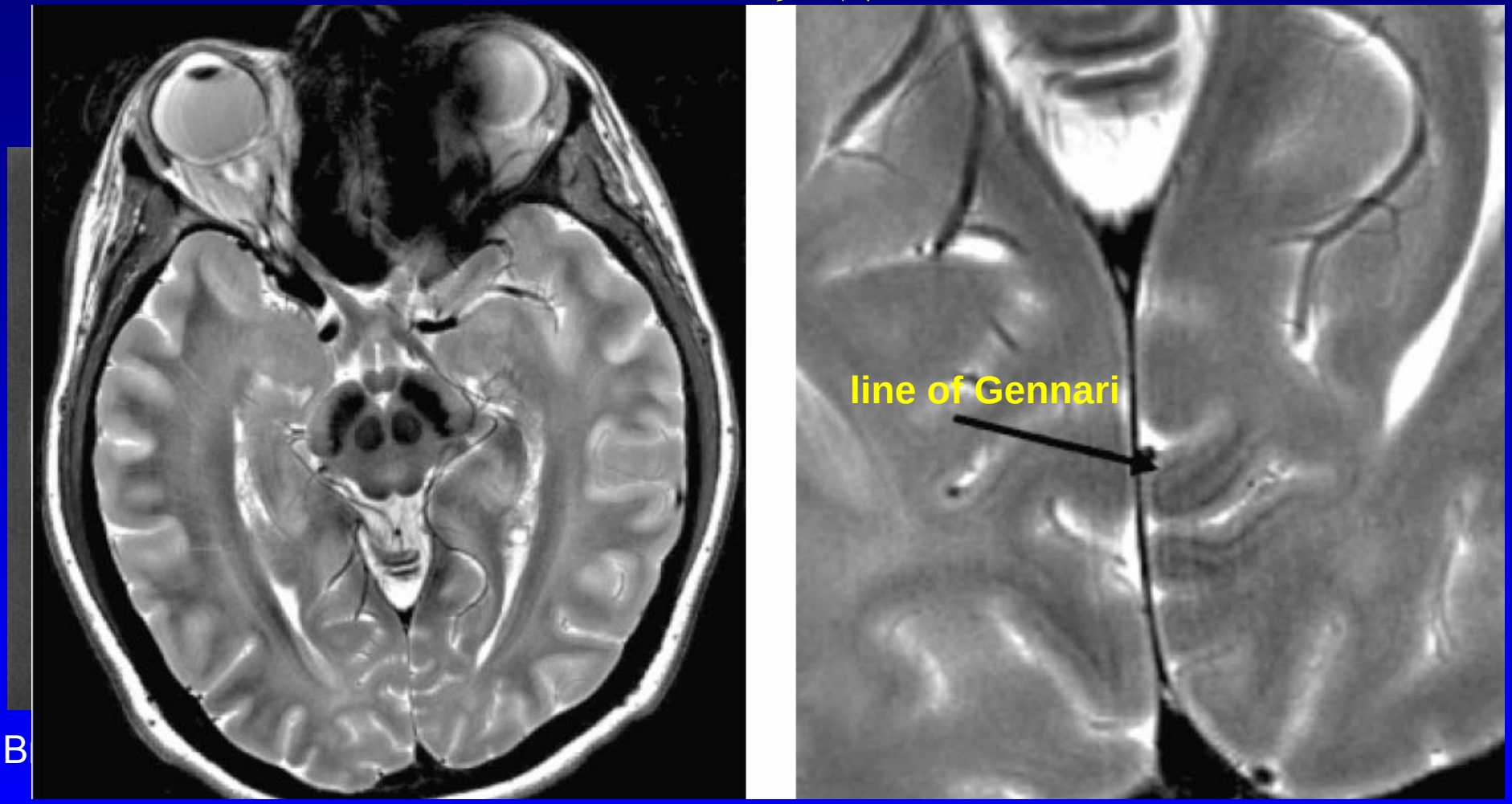
History of MRI

早期MR图像



Damadian (1977)

MRI 发展



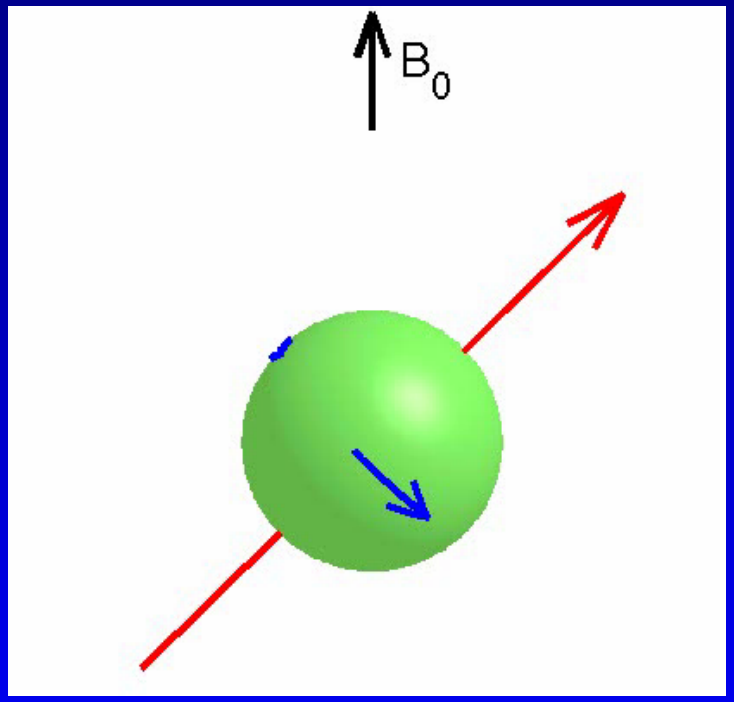
B

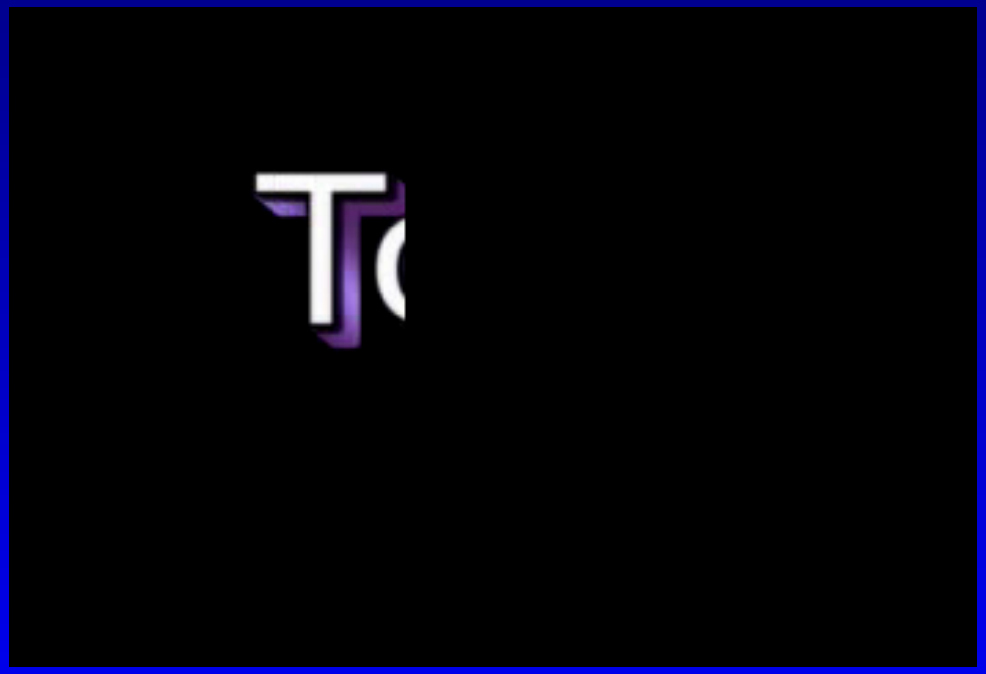
TSE, 512x760 matrix, (0.27x0.27x2mm), 9 min scan at 7T

1. van Uijen C.M.J et al. MRM 1984

磁共振基本原理

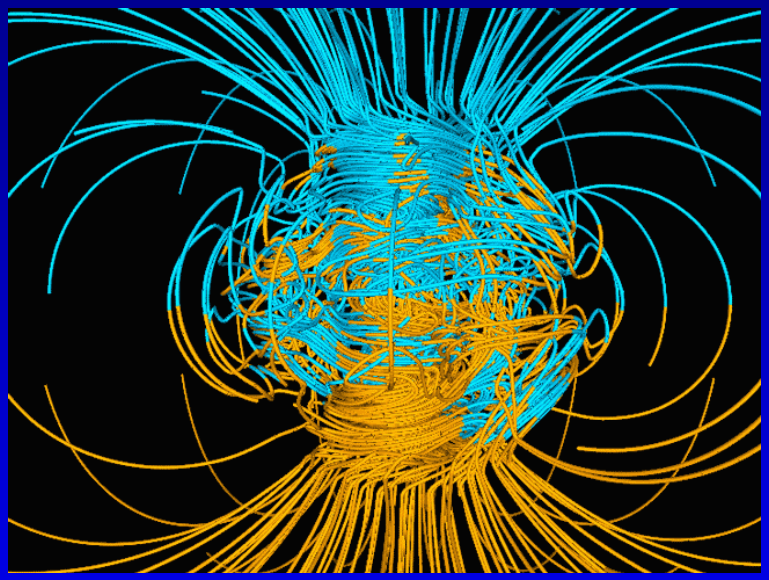
磁共振现象



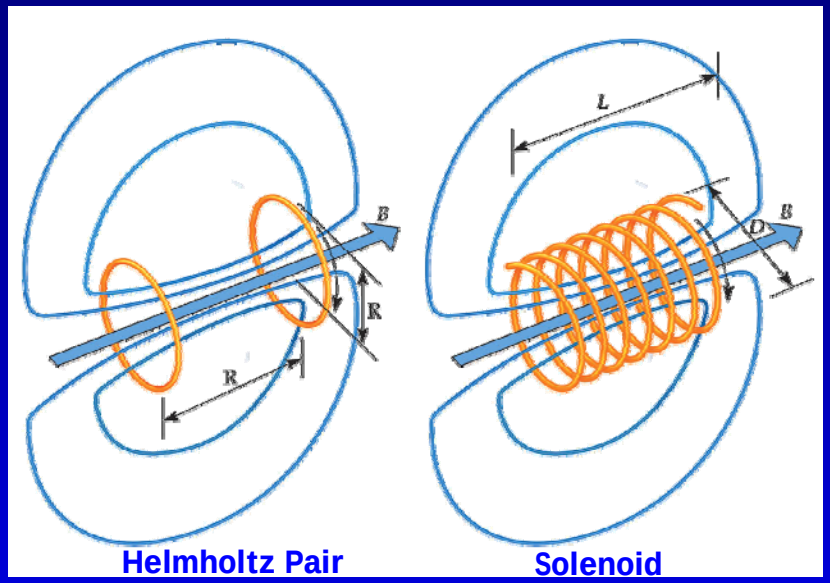


[质子的进动(precession)]

Principle of MRI
静主磁场



Computer simulation of Earth's magnetic field (~0.3-0.6 Gauss, or 6x10^{-5}T)



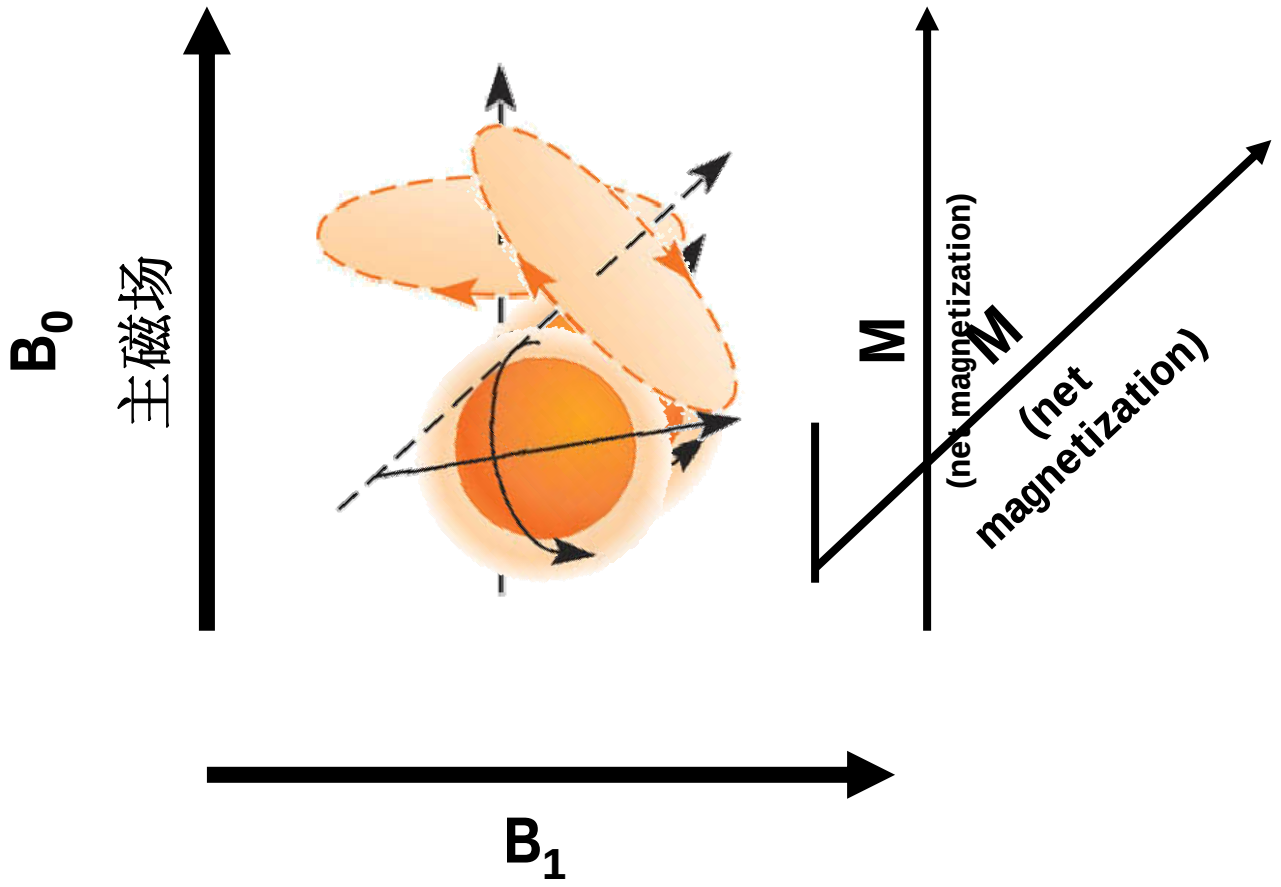
Helmholtz Pair Solenoid

B_0



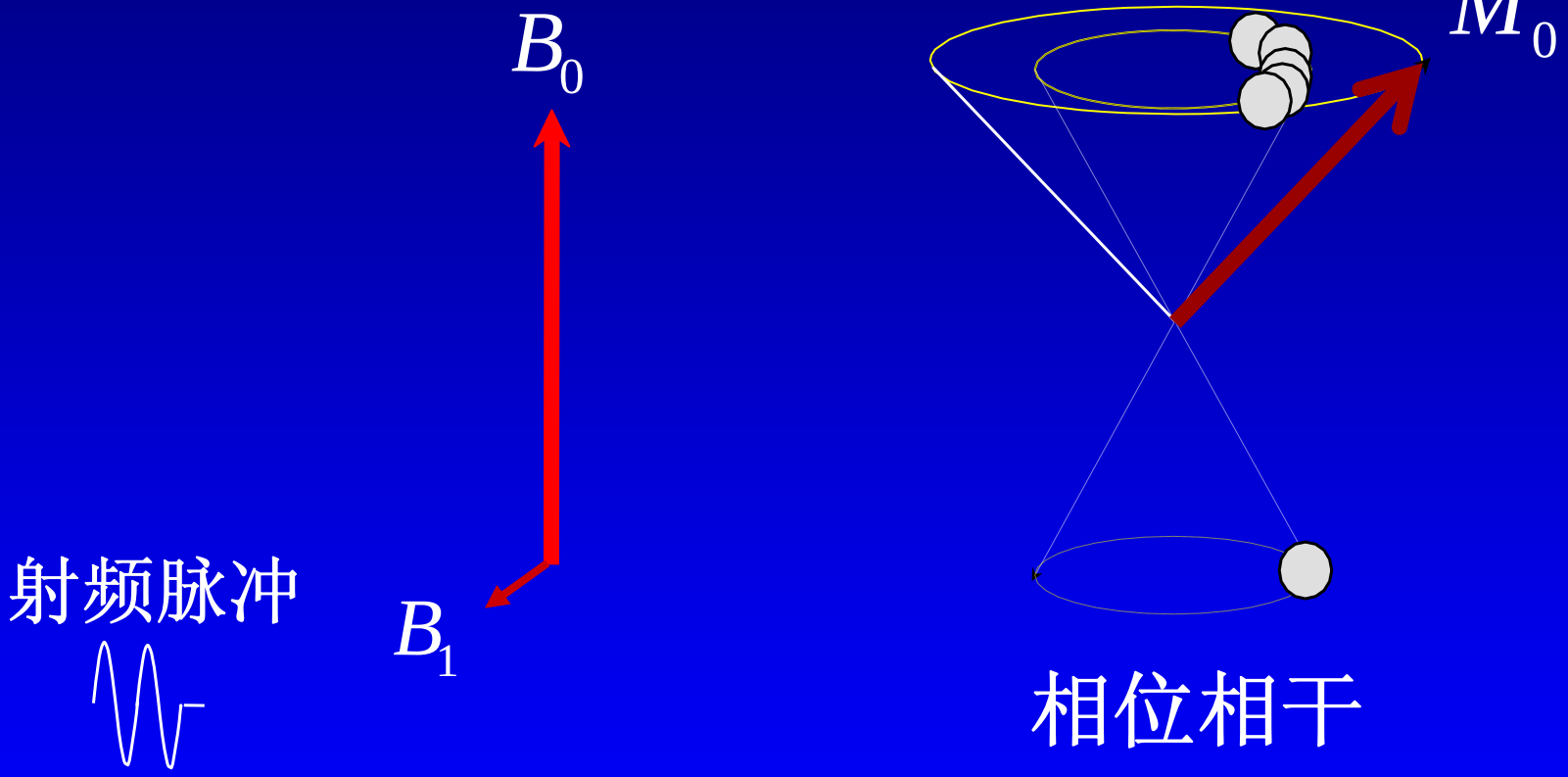
Philips 3T MRI Scanner

改变M的状态




This is a random illustrative photo. The internet is great.

磁共振现象



举例：　$B_0 = 1.5\,\mathrm{Tesla}$　　$B_1 = 50\,\mathrm{mT}$

信号的激发与接收

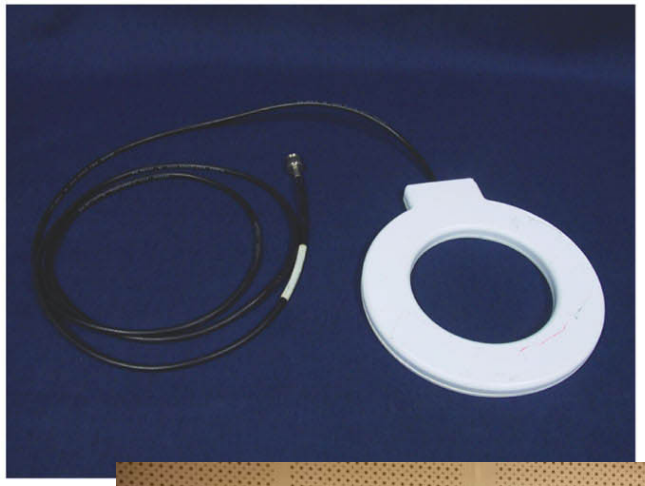
Excitation



Pippa Storey

信号的激发与接收：射频线圈

Surface Coil



Volume Coil

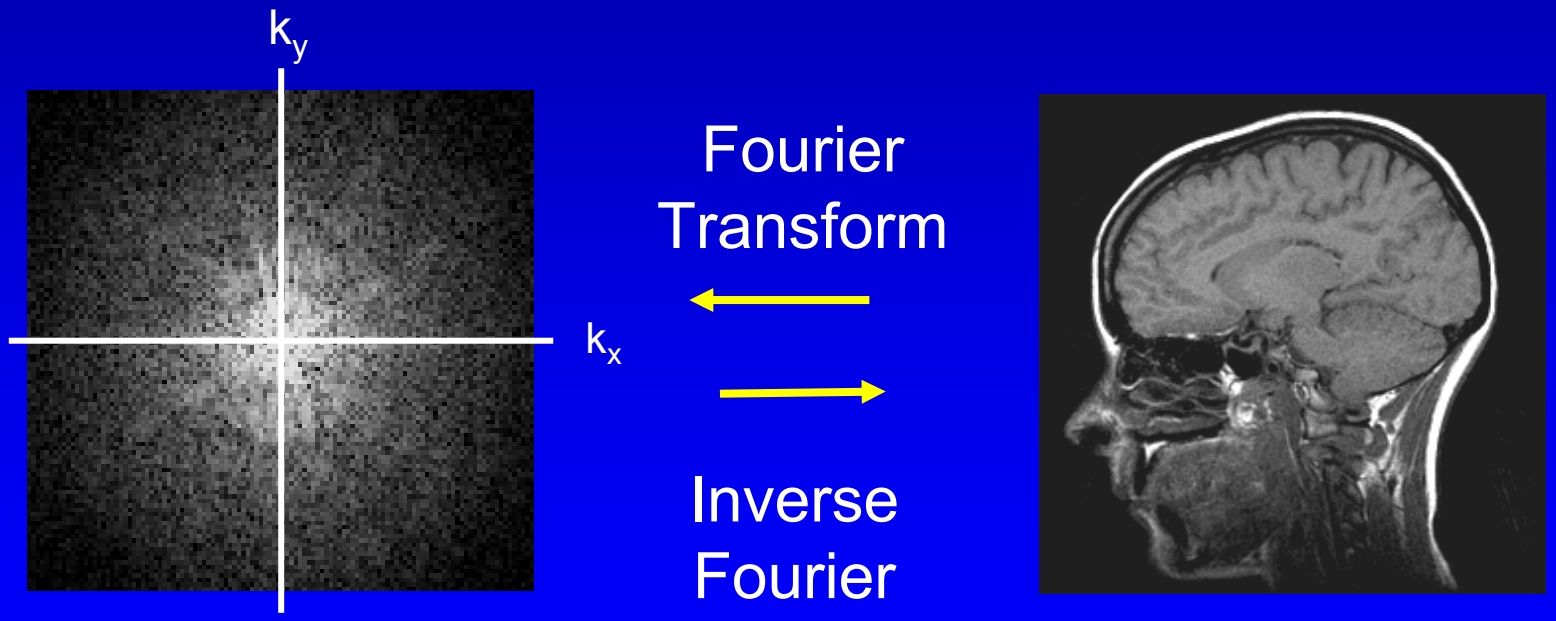




Electronic coils tuned to radio signals send energy into the brain and record an emitted “echo”.

图像生成

- k-space is the “spatial frequency” space
- Image data are acquired in k-space



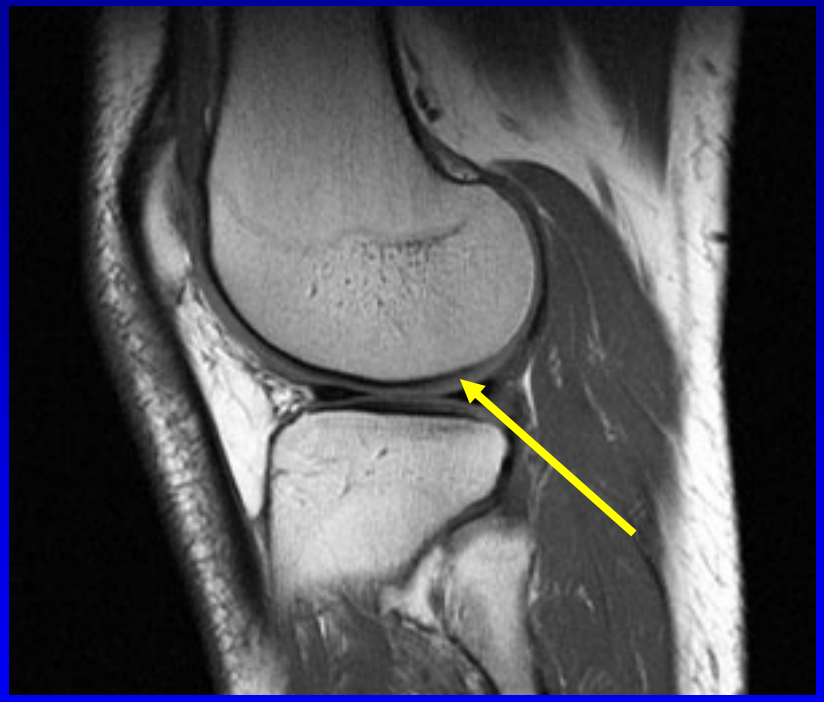
磁共振的特点

多姿多彩的磁共振
One picture is worth a thousand words

- 丰富的解剖结构信息



CT Image of Knee



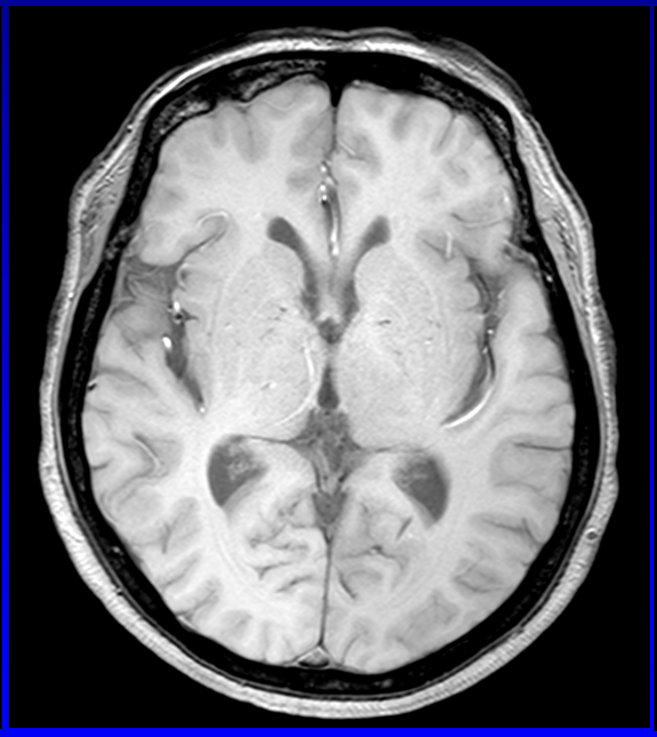
MRI Image of Knee

多姿多彩的磁共振
One picture is worth a thousand words

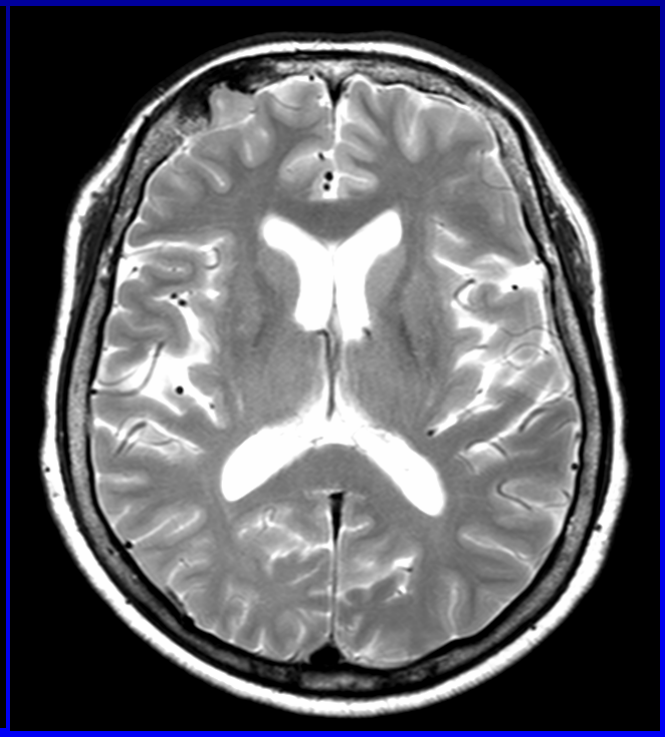
- 丰富的解剖结构信息



CT



MRI T1



MRI T2

多姿多彩的磁共振
One picture is worth a thousand words

- 丰富的解剖结构信息



3T高分辨率腕部成像

Ultra-high Resolution

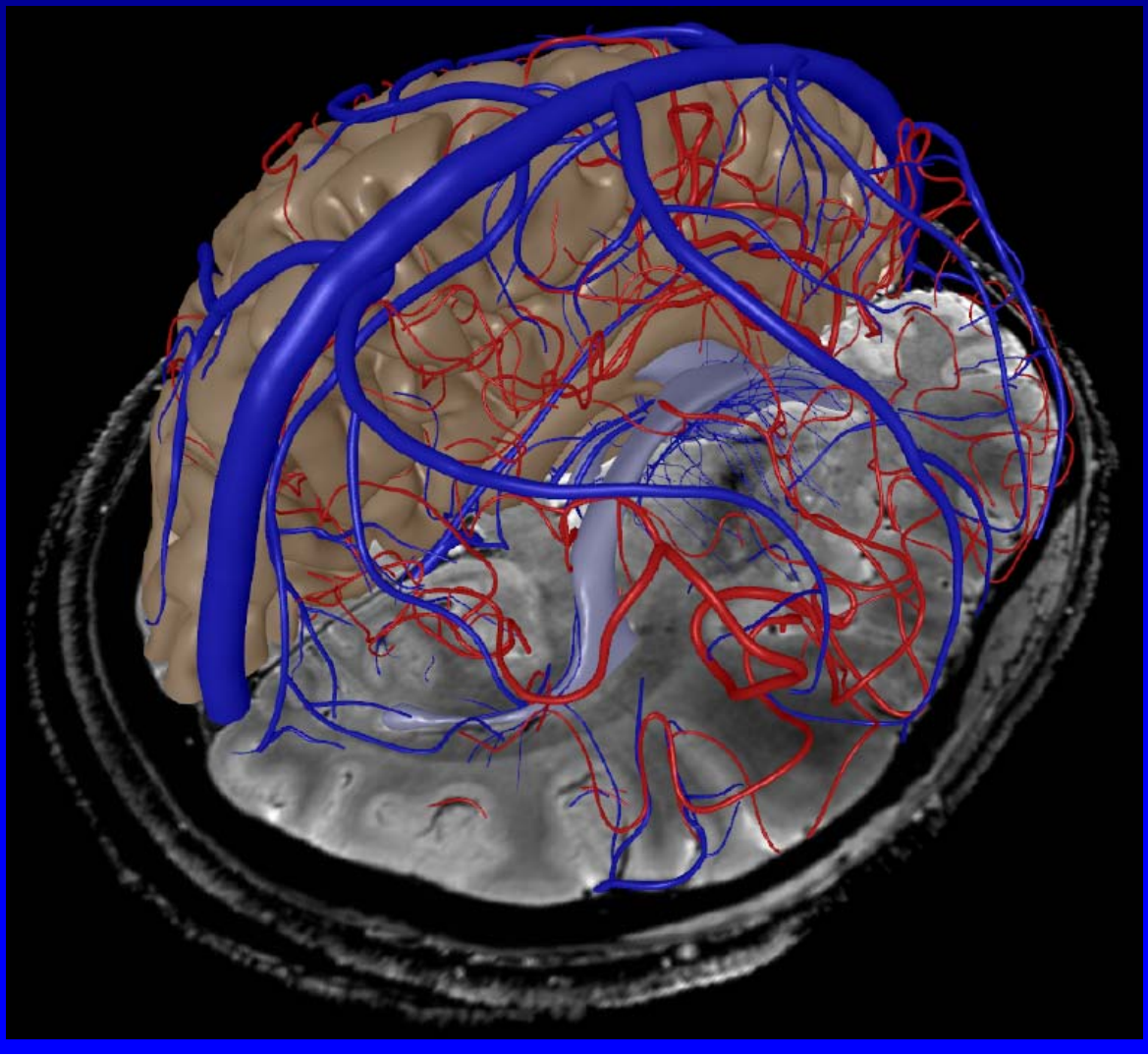
Wrist Imaging

PDW a-TSE
SENSE Wrist coil
SENSE x 2
16 slices x 2.0 mm
6:44 min

150 µm
resolution

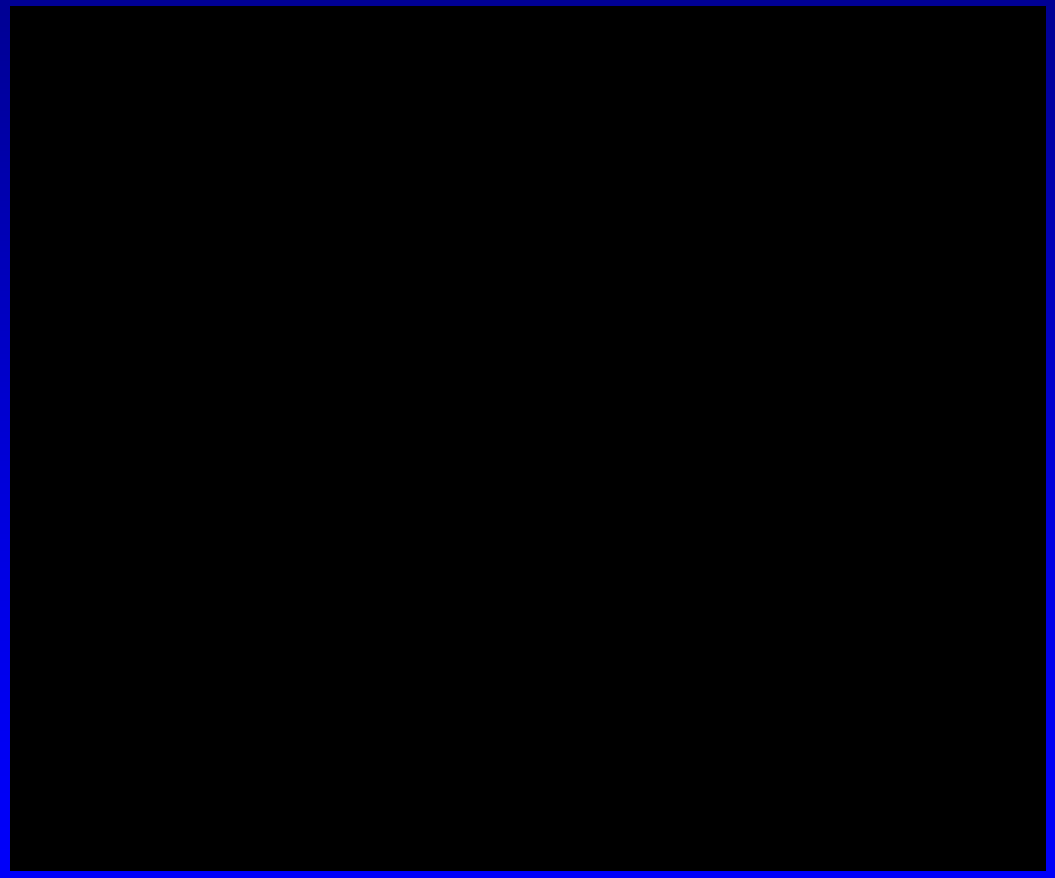
多姿多彩的磁共振
One picture is worth a thousand words

- 丰富的解剖结构信息



多姿多彩的磁共振
One picture is worth a thousand words

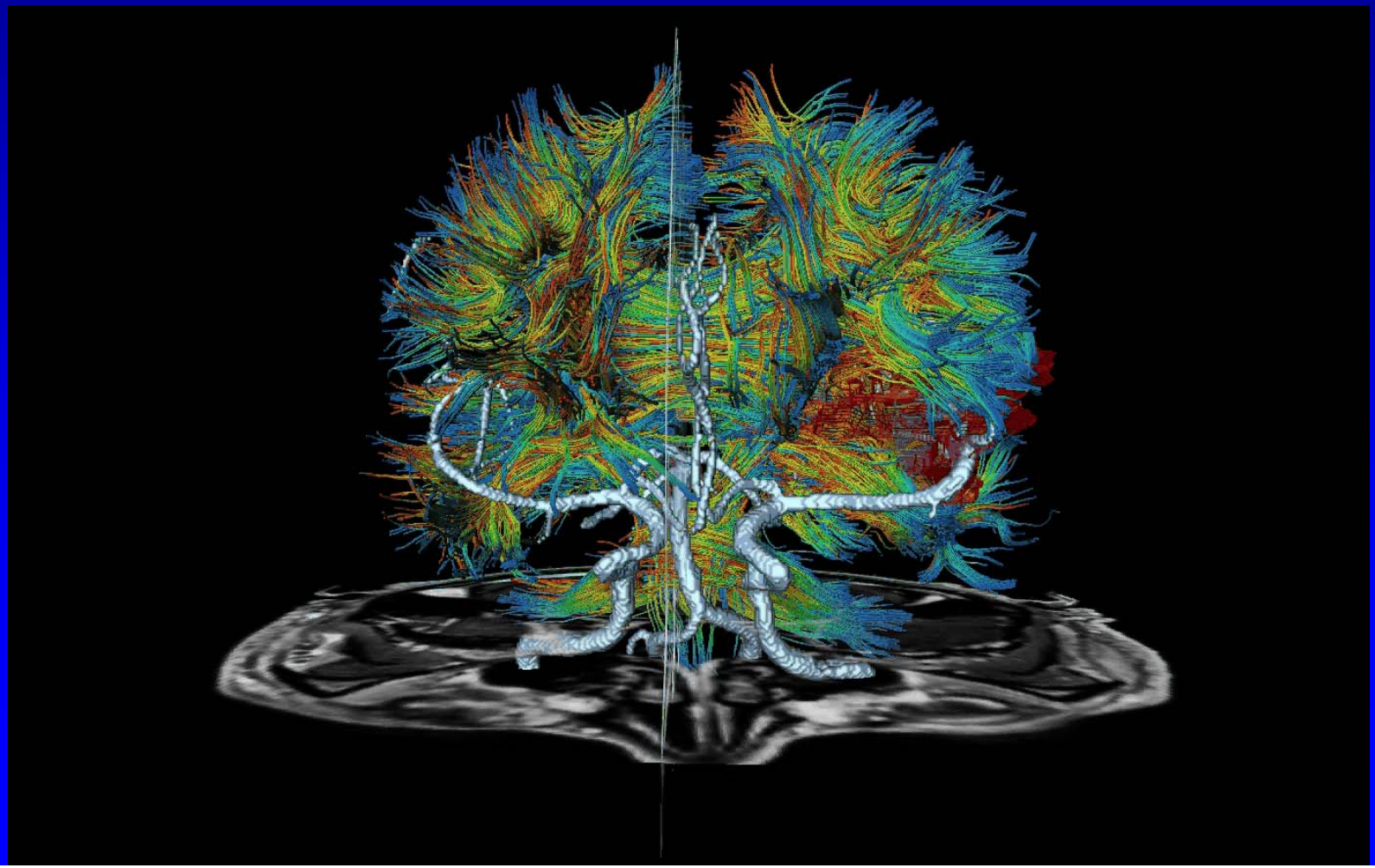
- 4-D信息



Courtesy of Dr. S Kozerke

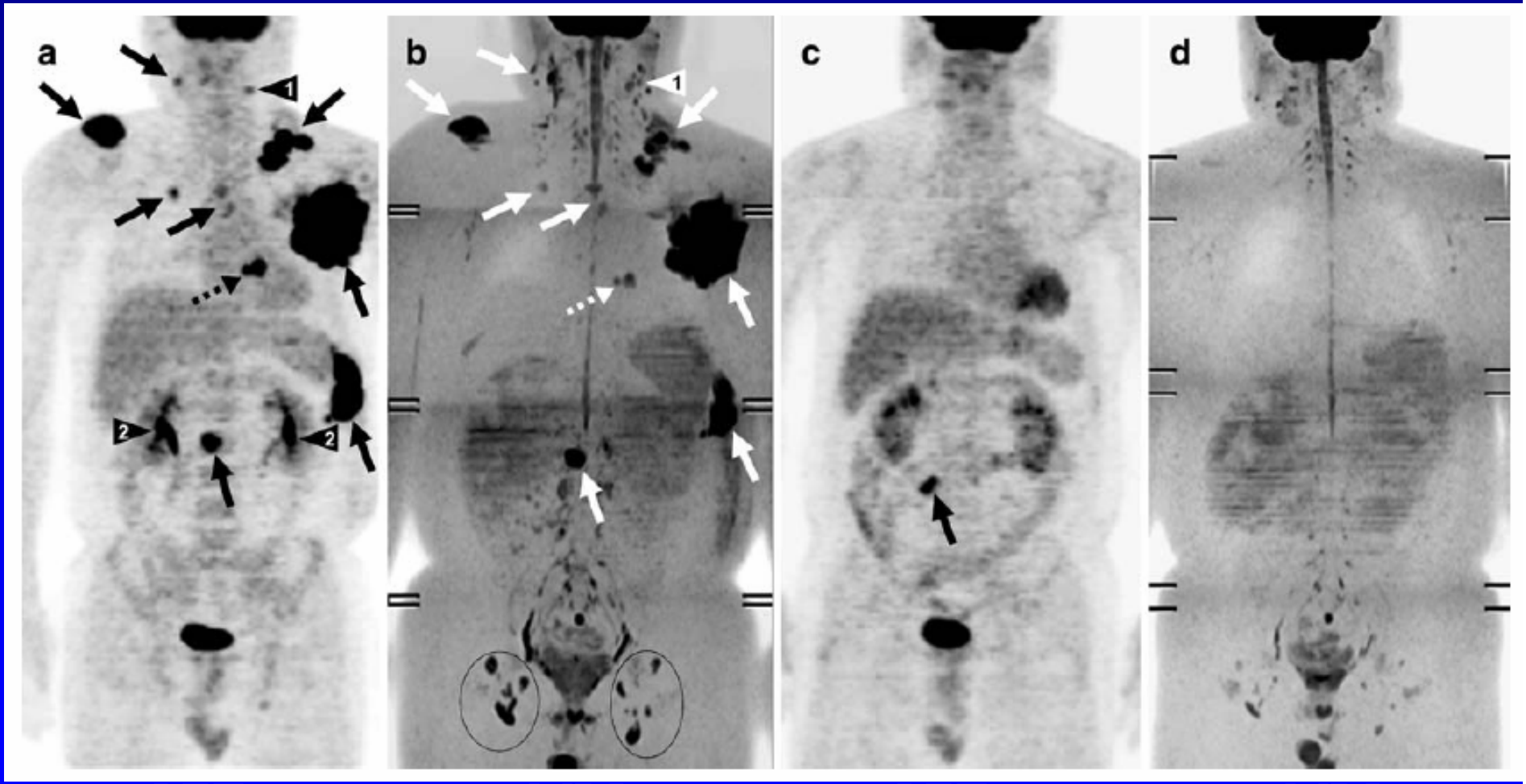
多姿多彩的磁共振
One picture is worth a thousand words

- 丰富的解剖结构和功能信息



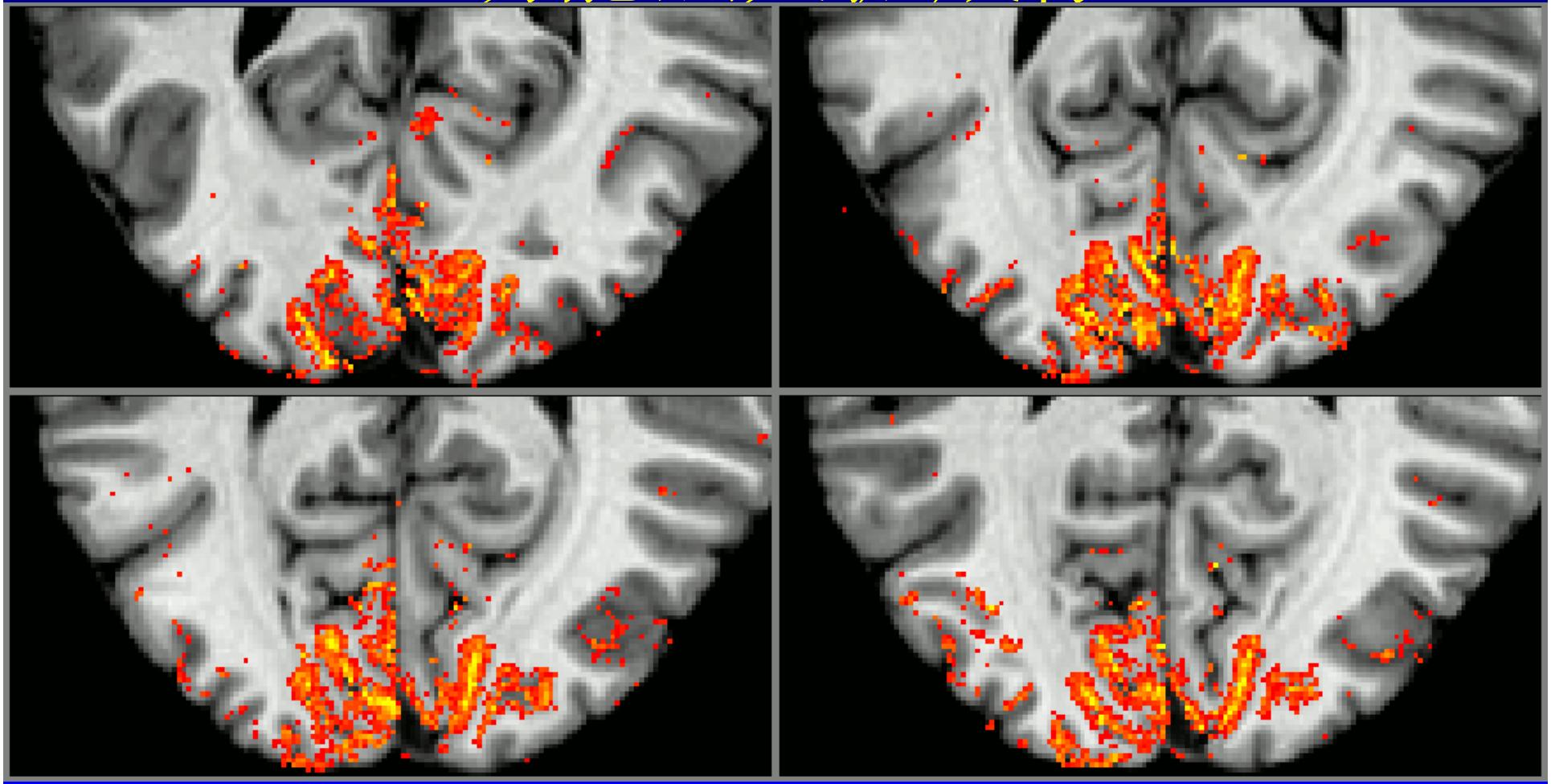
多姿多彩的磁共振
One picture is worth a thousand words

- 不断扩展的功能信息



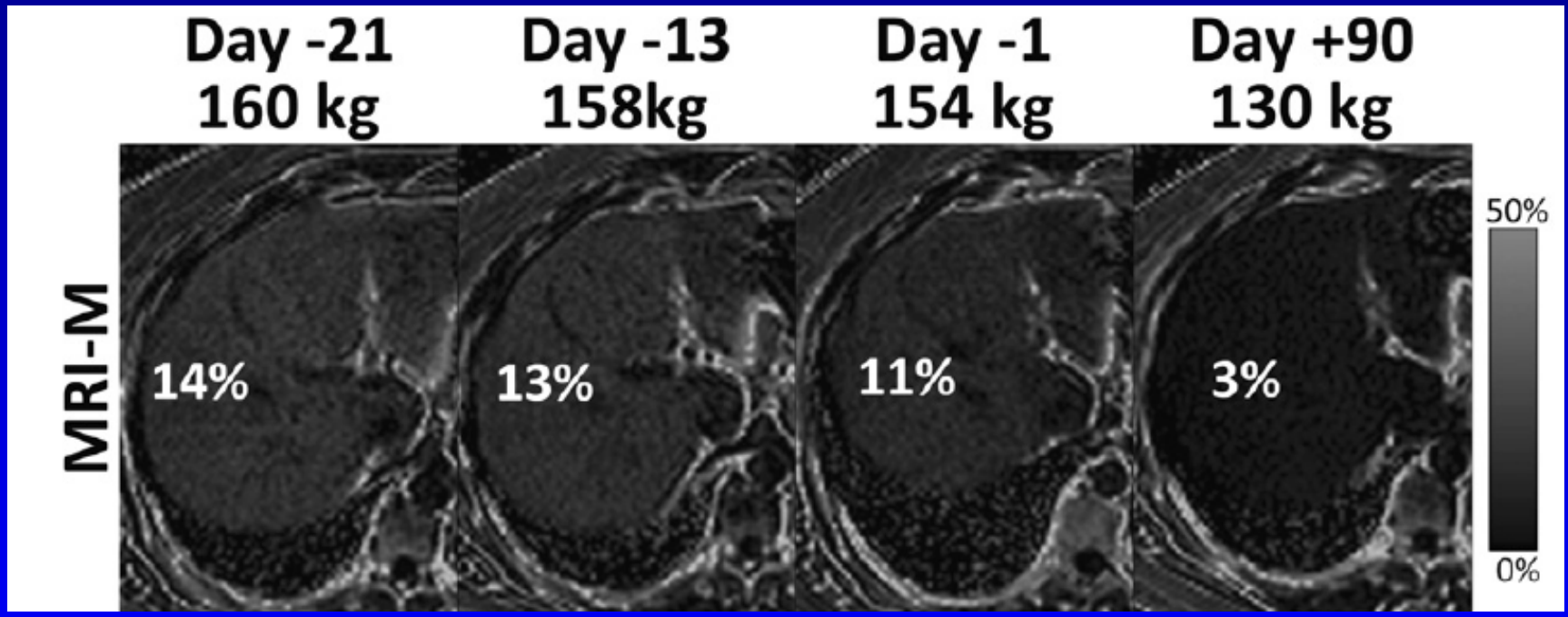
PET MRI PET MRI

functional MRI (fMRI)
功能磁共振成像



多姿多彩的磁共振
One picture is worth a thousand words

- 准确的定量信息



磁共振的特点 Be different in order to be seen

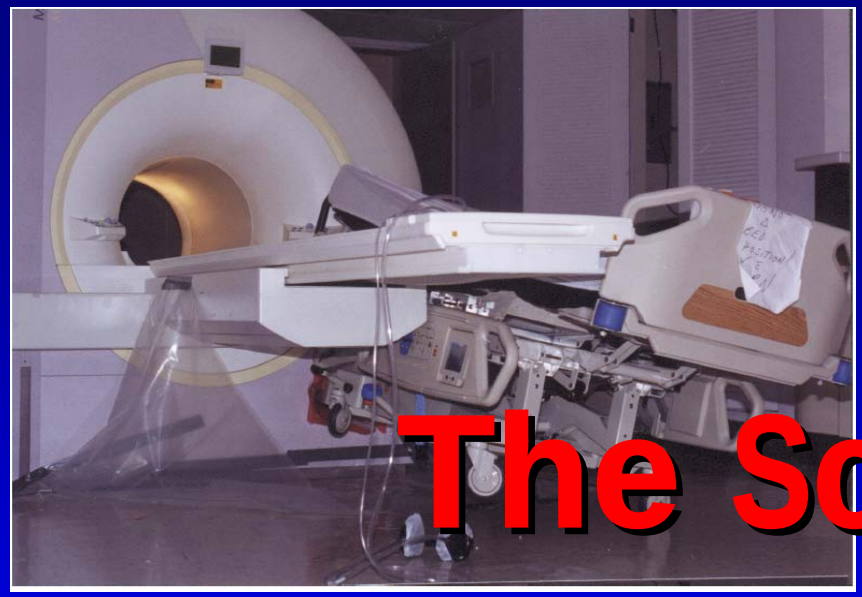
- 对人体无创伤、无电离辐射，安全；
- 优异的软组织对比度；
- 高度灵活的对比成像；
- 可以任意层面成像；
- 较高分辨率
- 提供功能信息

磁共振安全

- 已知急性副作用
  - Projectiles, rapid field changes, RF heating, claustrophobia, acoustic noise, etc.

- 慢性副作用的流行病学调查
  - Extended exposure to magnetic fields does not cause harm

投射效应





The Scanner is Never Off!





Take Home Message:

- MRI is safe
- MRI is fun

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Website: http://cbir.med.tsinghua.edu.cn/


Thank you